

Stronger Partial Resolution Packet for Problem 3.19

Run `fa.banach.001`: consolidated stronger partial

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Abstract

This packet upgrades the previous partial packet for Problem 3.19 of Rozendaal–Veraar, *Fourier multiplier theorems involving type and cotype*. It is still not a full intrinsic classification of all Banach pairs (X, Y) , but it adds several substantial results: the scalar classification, endpoint failures for every nonzero Banach pair, a corrected reading of Theorem 3.21, an endpoint correction to the positive-kernel statements, a closed-graph uniformization principle, monotonicity by fractional integration, an exact bilinear γ -Pitt criterion, a target cotype-index obstruction, and a complete classification for standard infinite-dimensional $L^u \rightarrow L^v$ and $\ell^u \rightarrow \ell^v$ pairs. The arbitrary Banach-pair classification remains open.

1 Source problem and status

The source paper is Jan Rozendaal and Mark Veraar, *Fourier multiplier theorems involving type and cotype*, arXiv:1605.09340; the arXiv page records the journal version in *J. Fourier Anal. Appl.* 24 (2018), 583–619 [2].

In Theorem 3.21 we provide conditions under which one can take $p = p_0$ and $q = q_0$. The general case we state as an open problem:

Problem 3.19. *Let $1 \leq p \leq 2 \leq q \leq \infty$ and $r \in (1, \infty]$ be such that $\frac{1}{r} = \frac{1}{p} - \frac{1}{q}$. Classify those Banach spaces X and Y for which $T_m \in \mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y))$ for all X -strongly measurable maps $m : \mathbb{R}^d \rightarrow \mathcal{L}(X, Y)$ such that $\{|\xi|^{d/r} m(\xi) : \xi \in \mathbb{R}^d \setminus \{0\}\}$ is γ -bounded.*

The same problem can be formulated in case m is scalar-valued, in which case

Problem 3.19 asks for a classification of Banach spaces X, Y for which, for

$$1 \leq p \leq 2 \leq q \leq \infty, \quad \frac{1}{r} = \frac{1}{p} - \frac{1}{q}, \quad r \in (1, \infty],$$

every X -strongly measurable symbol

$$m : \mathbb{R}^d \setminus \{0\} \rightarrow \mathcal{L}(X, Y)$$

whose weighted range

$$\{|\xi|^{d/r} m(\xi) : \xi \neq 0\}$$

is γ -bounded defines a bounded Fourier multiplier

$$T_m : L^p(\mathbb{R}^d; X) \rightarrow L^q(\mathbb{R}^d; Y).$$

We write this property as $P_{p,q}^d(X, Y)$, or simply $P_{p,q}(X, Y)$.

Status. This packet is a *strong partial result*. It gives complete answers in several important regimes, especially the scalar case, all endpoint faces, and the $L^u \rightarrow L^v$ scale. It does not give a familiar intrinsic classification for arbitrary Banach pairs.

2 Summary of strengthened conclusions

Topic	Result in this packet
Scalar case	$P_{p,q}(\mathbb{C}, \mathbb{C})$ holds exactly for $1 < p \leq 2 \leq q < \infty$.
Endpoint faces	If $X, Y \neq 0$, then $p = 1$ with finite $q \geq 2$ fails, and $q = \infty$ with $1 < p \leq 2$ fails.
Theorem 3.21	The positive lattice theorem is valid after deleting $p = 1$ and interpreting the target embedding as an isomorphic closed-subspace embedding.
Positive-kernel endpoint	The $p = 1 < q$ endpoint in the positive-kernel statements also fails by the scalar Riesz-potential obstruction.
Formal exact criterion	$P_{p,q}(X, Y)$ is equivalent to a bilinear γ -Pitt inequality.
Target geometry	If $X \neq 0$ and $P_{p,q}(X, Y)$ holds, then the cotype index of Y is at most q .
Lebesgue/sequence pairs	For standard infinite-dimensional $1 \leq u, v < \infty$, $P_{p,q}(L^u, L^v)$ holds exactly when $p \leq u$ and $v \leq q$.
Still open	A standard-invariant classification for arbitrary Banach pairs.

3 Scalar and endpoint classification

Theorem 3.1 (Scalar classification). *For $X = Y = \mathbb{C}$, the property $P_{p,q}(\mathbb{C}, \mathbb{C})$ holds if and only if*

$$1 < p \leq 2 \leq q < \infty.$$

Proof. For $1 < p \leq 2 \leq q < \infty$, scalar γ -boundedness is uniform boundedness, so

$$|m(\xi)| \leq C|\xi|^{-d/r}.$$

The function $|\xi|^{-d/r}$ belongs to weak L^r , and Hormander's weak- L^r multiplier theorem gives $L^p \rightarrow L^q$ boundedness [1]. The case $p = q = 2$ is Plancherel.

Let $p = 1$ and $2 \leq q < \infty$. The symbol

$$a_q(\xi) = |\xi|^{-d/q'}$$

has weighted range bounded. Its inverse Fourier transform is a nonzero multiple of $|x|^{-d/q}$. Testing T_{a_q} on a nonnegative approximate identity η_ε gives, on $2\varepsilon \leq |x| \leq 1/2$,

$$|T_{a_q}\eta_\varepsilon(x)| \gtrsim |x|^{-d/q},$$

and therefore

$$\|T_{a_q}\eta_\varepsilon\|_{L^q}^q \gtrsim \int_{2\varepsilon \leq |x| \leq 1/2} |x|^{-d} dx \simeq \log(1/\varepsilon) \rightarrow \infty.$$

The $q = \infty$ face follows by duality from the real even multiplier $|\xi|^{-d/p}$: boundedness $L^p \rightarrow L^\infty$ would imply boundedness $L^1 \rightarrow L^{p'}$, already contradicted. $\square$

Theorem 3.2 (Endpoint obstruction for nonzero pairs). *Let $X, Y \neq 0$. Then*

$$P_{1,q}(X, Y) \quad (2 \leq q < \infty) \quad \text{and} \quad P_{p,\infty}(X, Y) \quad (1 < p \leq 2)$$

both fail.

Proof. Choose $x_0 \in X$, $x^* \in X^*$, $y_0 \in Y$, and $y^* \in Y^*$ with $x^*(x_0) = y^*(y_0) = 1$. If a is one of the scalar counterexample symbols, put

$$m(\xi)x = a(\xi)x^*(x)y_0.$$

The weighted family is γ -bounded because, for $|\lambda_j| \leq 1$,

$$\left(\mathbb{E} \left\| \sum_j \gamma_j \lambda_j x^*(u_j) y_0 \right\|_Y^2 \right)^{1/2} \leq \|y_0\| \|x^*\| \left(\mathbb{E} \left\| \sum_j \gamma_j u_j \right\|_X^2 \right)^{1/2}.$$

Testing on $f = hx_0$ and applying y^* recovers the bad scalar estimate. □

4 Corrections to Theorem 3.21

each $j \in \mathcal{J}$, $\{|\xi|^{\frac{d}{r}} m_j(\xi) \mid \xi \in \mathbb{R}^d\} \subseteq \mathcal{L}(X, Y)$ is γ -bounded by C . Note that, since X and Y have finite cotype, γ -boundedness and R -boundedness are equivalent. Now we claim that $\{\widetilde{T}_{m_j} \mid j \in \mathcal{J}\} \subseteq \mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y))$ is γ -bounded as well. To prove this claim one can use the method of [22, Theorem 3.2]. Indeed, using their notation, it follows from the Kahane-Khintchine inequalities that $\text{Rad}(X)$ has the same type as X and $\text{Rad}(Y)$ has the same cotype as Y . Therefore, given $j_1, \dots, j_n \in \mathcal{J}$ and the corresponding $m_{j_1}, \dots, m_{j_n}$, one can apply Theorem 3.18 to the multiplier $M : \mathbb{R}^d \setminus \{0\} \rightarrow \mathcal{L}(\text{Rad}(X), \text{Rad}(Y))$ given as the diagonal operator with diagonal $(m_{j_1}, \dots, m_{j_n})$. In order to check the γ -boundedness one now applies property (α) as in [22, Estimate (3.2)].

3.5. Convexity, concavity and L^p - L^q results in lattices. In this section we will prove certain sharp results in p -convex and q -concave Banach lattices.

First of all, from the proof of Theorem 3.18 we obtain the following result with the sharp exponents p and q .

Theorem 3.21. *Let $p \in [1, 2]$, $q \in [2, \infty)$, and let $r \in [1, \infty]$ be such that $\frac{1}{r} = \frac{1}{p} - \frac{1}{q}$. Let X be a complemented subspace of a p -convex Banach lattice with finite cotype and Y a Banach space that is continuously embedded in a q -concave Banach lattice. Let $m : \mathbb{R}^d \rightarrow \mathcal{L}(X, Y)$ be an X -strongly measurable map such that $\{|\xi|^{\frac{d}{r}} m(\xi) \mid \xi \in \mathbb{R}^d \setminus \{0\}\} \subseteq \mathcal{L}(X, Y)$ is γ -bounded. Then T_m extends uniquely to a bounded map $\widetilde{T}_m \in \mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y))$ with*

$$(3.11) \quad \|\widetilde{T}_m\|_{\mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y))} \leq C \gamma(\{|\xi|^{\frac{d}{r}} m(\xi) \mid \xi \in \mathbb{R}^d \setminus \{0\}\}),$$

where C is a constant depending on X , Y , p , q and d .

Proof. In the case where X is a p -convex and Y is a q -concave Banach lattice, the

The source's Theorem 3.21 includes $p = 1$. This cannot be correct, already for $X = Y = \mathbb{C}$, by Theorem 3.1. The proof imports the critical embedding

$$\dot{H}_p^{d/p-d/2}(\mathbb{R}^d; X) \hookrightarrow \gamma(\mathbb{R}^d; X).$$

At $p = 1$ and $X = \mathbb{C}$, this would imply the false strong endpoint Sobolev embedding $\dot{H}_1^{d/2} \hookrightarrow L^2$. Thus the issue is not a normalization artifact; it is the classical failure of strong endpoint fractional integration.

There is also a target-space wording issue. The source assumes that Y is continuously embedded in a q -concave Banach lattice Y_0 . The proof obtains an estimate for the image $J_Y T_m f$ in Y_0 . To recover an estimate in Y , J_Y must be bounded below, i.e. an isomorphism onto a closed subspace. Mere injectivity is insufficient.

Proposition 4.1 (Mere target injection is insufficient). *There is a bounded injective map $J : \ell^v \rightarrow \ell^2$ for each $v > 2$, but $P_{2,2}(\mathbb{C}, \ell^v)$ fails. Hence Theorem 3.21 cannot be read with “continuous embedding” meaning only bounded injective map.*

Proof. The map $J(y_1, y_2, \dots) = (2^{-1}y_1, 2^{-2}y_2, \dots)$ is bounded and injective from ℓ^v to ℓ^2 . If this sufficed, the Hilbert target would give $P_{2,2}(\mathbb{C}, \ell^v)$. But Theorem 7.2 below, with $q = 2 < v$, gives a contradiction. $\square$

Theorem 4.2 (Corrected lattice theorem). *Let*

$$1 < p \leq 2 \leq q < \infty, \quad \frac{1}{r} = \frac{1}{p} - \frac{1}{q}.$$

Assume X is isomorphic to a complemented subspace of a p -convex Banach lattice X_0 with finite cotype, and Y is isomorphic to a closed subspace of a q -concave Banach lattice Y_0 . If

$$\{|\xi|^{d/r} m(\xi) : \xi \neq 0\}$$

is γ -bounded, then

$$T_m : L^p(\mathbb{R}^d; X) \rightarrow L^q(\mathbb{R}^d; Y)$$

is bounded.

Proof. For X_0, Y_0 , the valid lattice embeddings are

$$\dot{H}_p^{d/p-d/2}(\mathbb{R}^d; X_0) \hookrightarrow \gamma(\mathbb{R}^d; X_0), \quad \gamma(\mathbb{R}^d; Y_0) \hookrightarrow \dot{H}_q^{d/q-d/2}(\mathbb{R}^d; Y_0),$$

for $p > 1$. The source proof then applies with

$$m_1(\xi) = |\xi|^{d/2-d/p}, \quad m_2(\xi) = |\xi|^{d/r} m(\xi) m_1(\xi),$$

and the γ -multiplier theorem. For subspaces, use $\tilde{m}(\xi) = J_Y m(\xi) P_X$, where P_X is a bounded projection onto the copy of X and $J_Y : Y \rightarrow Y_0$ is bounded below. The Y_0 -estimate then transfers back to Y . $\square$

5 Another endpoint correction

The source’s Theorem 3.24 and Corollary 3.25 also include the face $p = 1 < q$. The same endpoint obstruction invalidates that face.

Proposition 5.1 (Positive-kernel endpoint failure). *Let $1 < q < \infty$, $s = d(1 - 1/q)$, and let $K \in C_c^\infty(\mathbb{R}^d)$ be nonnegative with $\int K = 1$. Convolution by K is not bounded*

$$\dot{H}_1^s(\mathbb{R}^d) \rightarrow L^q(\mathbb{R}^d).$$

Equivalently, the positive symbol $m_s(\xi) = |\xi|^{-s} \widehat{K}(\xi)$ does not define a bounded map $L^1 \rightarrow L^q$.

Proof. Let I_s be the Riesz potential, with kernel comparable to $|x|^{s-d} = |x|^{-d/q}$. For a nonnegative approximate identity η , $f = I_s \eta$ belongs to $\dot{H}_1^s$ with controlled norm. But

$$K * f = I_s(K * \eta).$$

Since $K * \eta$ is compactly supported and has integral one, its Riesz potential has asymptotic size $c|x|^{-d/q}$ at infinity, hence is not in L^q . $\square$

6 Uniformization and exact criterion

For the rest of the packet assume

$$1 < p \leq 2 \leq q < \infty, \quad a = d\left(\frac{1}{p} - \frac{1}{q}\right) < d.$$

Let $\mathcal{G}(X, Y)$ be the Banach space of essentially γ -bounded strongly measurable maps $M : \mathbb{R}^d \rightarrow \mathcal{L}(X, Y)$, with norm the essential γ -bound of the range.

Theorem 6.1 (Closed-graph uniformization). *If $P_{p,q}(X, Y)$ holds qualitatively, then there is C such that for every $M \in \mathcal{G}(X, Y)$,*

$$\|T_{|\cdot|^{-a}} M\|_{\mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y))} \leq C \|M\|_{\mathcal{G}}.$$

Proof. The map

$$\Phi : \mathcal{G}(X, Y) \rightarrow \mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y)), \quad \Phi(M) = T_{|\cdot|^{-a}} M,$$

is everywhere defined by the qualitative property. If $M_n \rightarrow M$ in $\mathcal{G}$ and $\Phi(M_n) \rightarrow S$ in operator norm, then for finite-rank Schwartz f the functions $\Phi(M_n)f$ converge in distributions to $\Phi(M)f$, because $a < d$ and $\hat{f}$ is rapidly decreasing. They also converge in distributions to Sf . Hence $S = \Phi(M)$ on a dense subspace, so the graph is closed. The closed graph theorem gives the bound. $\square$

Theorem 6.2 (Monotonicity). *If*

$$1 < p \leq p_0 \leq 2 \leq q_0 \leq q < \infty$$

and $P_{p_0, q_0}(X, Y)$ holds, then $P_{p,q}(X, Y)$ holds.

Proof. Put

$$\alpha = d\left(\frac{1}{p} - \frac{1}{p_0}\right), \quad \beta = d\left(\frac{1}{q_0} - \frac{1}{q}\right).$$

For a symbol m at (p, q) , set $n(\xi) = |\xi|^{\alpha+\beta} m(\xi)$. The weighted families for m at (p, q) and n at (p_0, q_0) are identical. Since

$$T_m = I_\beta T_n I_\alpha,$$

and vector-valued Hardy–Littlewood–Sobolev estimates for the positive Riesz kernels hold for every Banach space, the result follows. $\square$

For finite-rank Schwartz $f : \mathbb{R}^d \rightarrow X$ and $g : \mathbb{R}^d \rightarrow Y^*$, define

$$\Pi_a^\gamma(f, g) = \sup_{\|M\|_{\mathcal{G}} \leq 1} \left| \int_{\mathbb{R}^d} |\xi|^{-a} \langle M(\xi) \hat{f}(\xi), \hat{g}(-\xi) \rangle d\xi \right|.$$

Theorem 6.3 (Exact bilinear γ -Pitt criterion). *For nonzero X, Y , $P_{p,q}(X, Y)$ holds if and only if there is C such that*

$$\Pi_a^\gamma(f, g) \leq C \|f\|_{L^p(\mathbb{R}^d; X)} \|g\|_{L^{q'}(\mathbb{R}^d; Y^*)}$$

for all finite-rank Schwartz f, g .

Proof. The forward implication is Theorem 6.1 and Fourier-side duality. Conversely, for $M \in \mathcal{G}(X, Y)$ and finite-rank Schwartz f , the inverse transform $h = T_{|\cdot|^{-a}M}f$ is a continuous Y -valued function. The displayed bilinear inequality bounds the pairing of h with dense finite-rank $L^{q'}(Y^*)$ test functions. The standard Bochner norming argument gives

$$\|h\|_{L^q(Y)} \leq C \|M\|_{\mathcal{G}} \|f\|_{L^p(X)}.$$

Density gives the multiplier extension. □

Remark 6.4. *The criterion is exact but not an intrinsic classification in familiar Banach space terms. This is why the packet remains partial for the original Problem 3.19.*

7 Frequency-translation obstructions

Fix nonzero $k, \phi \in C_c^\infty(\mathbb{R}^d)$, set $\psi = \widehat{k}$, and put $h = k * \phi \neq 0$. Choose translations $b_1, \dots, b_n$ so far apart that the relevant translates of ϕ and h are disjoint. Since ψ is Schwartz,

$$\sup_{\xi \neq 0} |\xi|^a |\psi(\xi)| < \infty.$$

The phases $e^{\pm 2\pi i b_j \cdot \xi}$ make separated inputs add coherently after multiplication in frequency.

Theorem 7.1 (Domain obstruction). *Let $1 \leq u < p$, let $Y \neq 0$, and suppose X contains uniformly complemented copies of ℓ_n^u . Then $P_{p,q}(X, Y)$ fails.*

Proof. It suffices to use the isometric model $X = \ell^u$. Let e_j be the first basis vectors and define

$$R_\theta x = n^{-1/u'} \sum_{j=1}^n \theta_j e_j^*(x), \quad \theta \in \mathbb{T}^n.$$

Because $u < p \leq 2$, $u < 2$, and the family $\{R_\theta\} \subset \mathcal{L}(\ell^u, \mathbb{C})$ is uniformly γ -bounded. Fix $y_0 \in Y$, $\|y_0\| = 1$, and set

$$M_n(\xi)x = n^{-1/u'} \sum_{j=1}^n e^{2\pi i b_j \cdot \xi} e_j^*(x) y_0, \quad m_n(\xi) = \psi(\xi) M_n(\xi).$$

The weighted γ -bounds are uniform. For

$$f_n(t) = \sum_{j=1}^n \phi(t - b_j) e_j,$$

disjointness gives $\|f_n\|_{L^p(\ell^u)} = n^{1/p} \|\phi\|_p$, while phase cancellation gives

$$T_{m_n} f_n = n^{1/u} h y_0.$$

Uniformization would force $n^{1/u} \lesssim n^{1/p}$, impossible when $u < p$. □

Theorem 7.2 (Target obstruction). *Let $q < v < \infty$, let $X \neq 0$, and suppose Y contains uniformly isomorphic copies of ℓ_n^v . Then $P_{p,q}(X, Y)$ fails.*

Proof. Choose embeddings $J_n : \ell_n^v \rightarrow Y$ with uniformly bounded distortion. Fix $x_0 \in X$, $x^* \in X^*$, $x^*(x_0) = 1$. For $\theta \in \mathbb{T}^n$, define

$$S_\theta z = n^{-1/v} z J_n \left(\sum_{j=1}^n \theta_j e_j \right).$$

The family $\{S_\theta\} \subset \mathcal{L}(\mathbb{C}, Y)$ is uniformly γ -bounded, using the type 2 behavior of fixed L^v -geometry for $v > 2$. Put

$$M_n(\xi)x = n^{-1/v} x^*(x) J_n \left(\sum_{j=1}^n e^{-2\pi i b_j \cdot \xi} e_j \right), \quad m_n = \psi M_n.$$

For $f = \phi x_0$,

$$T_{m_n} f(t) = n^{-1/v} J_n \left(\sum_{j=1}^n h(t - b_j) e_j \right),$$

so disjointness gives

$$\|T_{m_n} f\|_{L^q(Y)} \gtrsim n^{1/q-1/v} \|h\|_q \rightarrow \infty,$$

while $\|f\|_{L^p(X)}$ is fixed. □

Theorem 7.3 (ℓ^∞ target obstruction). *If $X \neq 0$ and Y contains ℓ_n^∞ uniformly, then $P_{p,q}(X, Y)$ fails for every finite q .*

Proof. Use the preceding construction with

$$c_n = (1 + \log(n+1))^{-1/2}$$

in place of $n^{-1/v}$. The Gaussian maximal inequality gives a uniform γ -bound for the phase family into ℓ_n^∞ , while the output norm grows like

$$\frac{n^{1/q}}{\sqrt{\log(n+1)}}.$$

□

Let

$$q_Y = \inf\{s \in [2, \infty) : Y \text{ has cotype } s\},$$

with $q_Y = \infty$ if Y has no finite cotype.

Corollary 7.4 (Target cotype-index obstruction). *If $X \neq 0$ and $P_{p,q}(X, Y)$ holds, then $q_Y \leq q$. In particular Y must have finite cotype.*

Proof. By the Maurey–Pisier finite representability theorem, if $q_Y > q$, then ℓ_n^v for some $v > q$, or ℓ_n^∞ , is uniformly finitely representable in Y . Apply Theorem 7.2 or Theorem 7.3. □

8 Complete $L^u \rightarrow L^v$ classification

Theorem 8.1 (Infinite-dimensional Lebesgue pairs). *Let $1 \leq u, v < \infty$, and let $L^u(S)$, $L^v(T)$ be infinite-dimensional Lebesgue spaces over measure spaces admitting arbitrarily many pairwise disjoint sets of finite positive measure. For*

$$1 < p \leq 2 \leq q < \infty,$$

one has

$$P_{p,q}(L^u(S), L^v(T)) \iff p \leq u \text{ and } v \leq q.$$

Proof. If $p \leq u$, then L^u is p -convex and has finite cotype. If $v \leq q$, then L^v is q -concave. The positive direction follows from Theorem 4.2. If $u < p$, disjoint normalized indicators span 1-complemented ℓ_n^u 's in L^u , so Theorem 7.1 applies. If $v > q$, disjoint normalized indicators give uniformly isomorphic ℓ_n^v 's in L^v , so Theorem 7.2 applies. $\square$

Corollary 8.2 (Sequence spaces). *For $1 \leq u, v < \infty$,*

$$P_{p,q}(\ell^u, \ell^v) \iff p \leq u \text{ and } v \leq q.$$

In particular,

$$P_{p,q}(\ell^u, \ell^u) \iff p \leq u \leq q.$$

9 Remaining open core

The results above strongly constrain the answer and solve many concrete regimes, but they do not finish the original problem for arbitrary Banach pairs. The main remaining task is to turn the exact γ -Pitt criterion of Theorem 6.3 into a classification in standard Banach-space invariants.

Problem 9.1. *Classify the pairs (X, Y) for which the bilinear criterion in Theorem 6.3 holds using familiar invariants such as type, cotype, Fourier type, K -convexity, or local finite representability.*

Problem 9.2. *Decide whether $P_{p,q}(X, Y)$ forces type p of X or cotype q of Y at the exact endpoint without the extra assumptions in known rank-one Pitt converse arguments.*

Problem 9.3. *Classify $P_{p,q}(L^\infty(S), Y)$ for finite-cotype targets Y . The corrected lattice theorem does not apply because L^∞ lacks finite cotype.*

10 Novelty and verification notes

The source paper, local run index, and web/arXiv searches through June 18, 2026 were checked for Problem 3.19, Theorem 3.21, errata/corrigenda, weighted γ -bounded multiplier ranges, and $L^u \rightarrow L^v$ or $\ell^u \rightarrow \ell^v$ formulations. No official correction to the $p = 1$ endpoint and no matching published $L^u \rightarrow L^v$ cross-pair classification were located. This is bounded evidence, not an exhaustive novelty proof.

Human review should focus on:

- (i) the closed-graph argument for $\mathcal{G}(X, Y)$;
- (ii) the closed-range interpretation of the target embedding in Theorem 3.21;
- (iii) the positive-kernel endpoint counterexample;
- (iv) the uniform γ -bounds in the frequency-translation constructions;
- (v) the finite-representability step in Corollary 7.4.

References

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